Design Document: Detecting Out-of-Context Image-Caption Misinformation

DATA5500 Senior Design | Fall 2026 | Instructor: Weijie Pang | Student: Mohamed Mohamed | September 27, 2026

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1. Overview of the Project, Including Its Potential Impact and Market

This project builds a system that detects out-of-context (OOC) misinformation: cases where a real, unaltered photograph is paired with a caption that misrepresents what it actually shows. Using CLIP, a pretrained model that embeds images and text into a shared vector space, the system measures whether a given image and caption are semantically consistent, then classifies the pair as a likely genuine match or a likely out-of-context mismatch. The project spans the full pipeline: acquiring and structuring benchmark data, generating embeddings, training and comparing classifiers of increasing sophistication, evaluating performance by mismatch type, and deploying the result as a live, interactive application.

The potential impact reaches several audiences. Fact-checking organizations and journalists currently rely on manual investigation or general-purpose reverse-image search to catch this specific misinformation pattern; a lightweight, automated first-pass signal could help them triage claims faster. Social media platforms and content moderation teams face the same problem at a much larger scale, where even a partial automated signal has real value. There is also a media literacy and education angle: a public-facing demo makes an otherwise abstract concept, that a caption can lie even when a photo is completely real, immediately tangible to a general audience. Within the scope of this course, the project also serves as a portfolio piece demonstrating an end-to-end multimodal machine learning system.

2. Existing Datasets, Resources, Tools, and Infrastructure

Datasets

- **NewsCLIPpings** (github.com/g-luo/news_clippings, also mirrored at huggingface.co/g-luo/news-clippings): the primary benchmark, providing labeled genuine and falsified image-caption pairs across defined mismatch-generation-method subsets (person-match, scene-match, and semantic/CLIP-based match).
- **VisualNews** (github.com/FuxiaoLiu/VisualNews-Repository): the underlying large-scale news image-caption corpus that NewsCLIPpings draws from; needed to resolve and download the actual source images referenced by its annotation files.
- **COSMOS** (github.com/shivangi-aneja/COSMOS): a related out-of-context misinformation benchmark, planned as an optional secondary check on whether findings generalize beyond NewsCLIPpings' specific mismatch-generation methodology.

Tools

- Python for the full pipeline: data handling, embedding generation, modeling, and the application layer.
- OpenAI's CLIP model, accessed via Hugging Face Transformers or the official openai/CLIP repository, used off-the-shelf through transfer learning rather than trained from scratch.
- scikit-learn for the logistic regression baseline and standard evaluation metrics; PyTorch for the lightweight neural network classifier.
- Streamlit for the deployed, interactive demo application.
- Matplotlib and/or Plotly for the evaluation dashboard's charts and the embedding-space visualization.
- Jupyter notebooks / Google Colab for exploratory development and experimentation.

Infrastructure

- Local machine for day-to-day development, scripting, and running the Streamlit app during testing.

- Google Colab's free GPU tier for the embedding-generation step, which is the most compute-intensive part of the pipeline.
- GitHub for version control and as the reproducible record of the project's code and documentation.
- A public hosting option for the live demo (Streamlit Community Cloud or Hugging Face Spaces) is under consideration; see Pending Design Decisions.

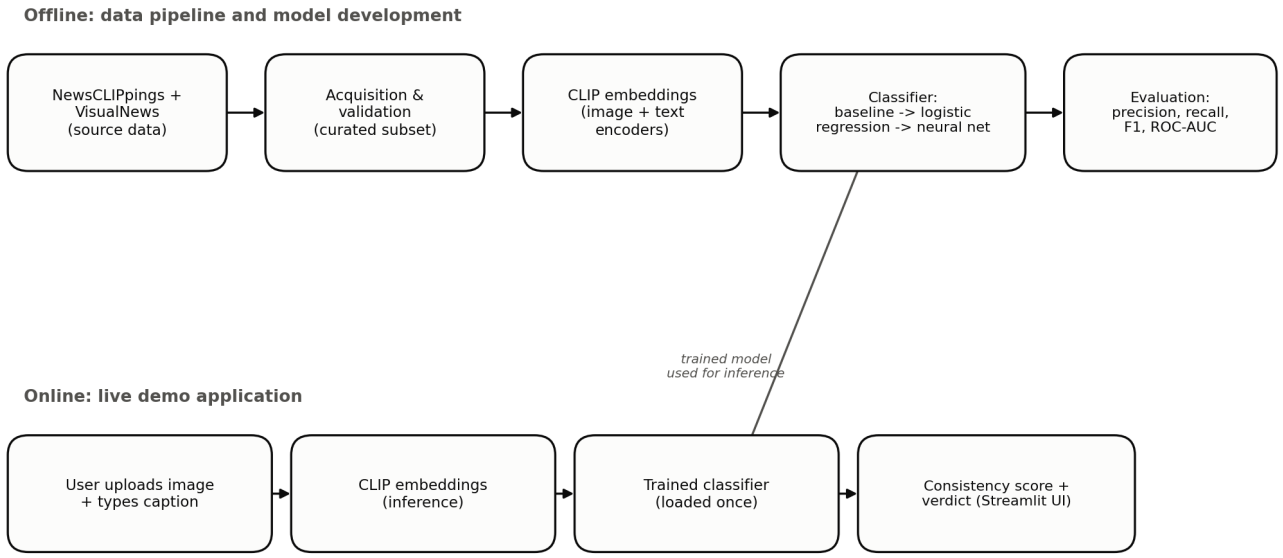


Figure 1. System architecture and data flow. The top row is the offline pipeline that produces a trained classifier; the bottom row is the online path a user's submission takes through the deployed application, using that same trained classifier for inference.

3. Key Design Decisions Already Made

Detection approach: embedding similarity over pixel forensics

Because out-of-context misinformation uses a genuine, unaltered image, the project is built around comparing image and caption meaning (via CLIP embeddings) rather than looking for pixel-level manipulation, which would find nothing wrong with these images by design.

Primary dataset: NewsCLIPpings, not custom scraping

NewsCLIPpings was chosen as the core dataset instead of scraping live news or social media content. This avoids the acquisition reliability and terms-of-service risk that comes with scraping, and provides an established, labeled benchmark with a defined evaluation methodology.

Modeling approach: staged, transfer-learning based

The classifier is being developed in stages: a similarity-threshold baseline, then a logistic regression model, then a lightweight neural network, all built on top of pretrained CLIP embeddings rather than training a vision-language model from scratch. This keeps the technical scope realistic for one student in one semester.

Evaluation approach: per-category breakdown as a first-class result

Rather than reporting a single aggregate accuracy figure, the evaluation plan treats performance by mismatch-generation method (person-match, scene-match, and semantic/CLIP-based match) as a primary result, since these methods are expected to differ meaningfully in difficulty.

Deployment approach: Streamlit with preset examples plus user upload

Streamlit was chosen for the live demo because it allows a working interactive interface to be built quickly by a solo developer. The interface will support both user-uploaded images and a small set of preset example pairs, so the demo works even for a visitor without an image on hand.

Scope boundary: a decision-support signal, not a verdict

The tool will be framed and documented as a decision-support signal (a consistency score with an explanation), not as a definitive true/false judgment, to avoid overstating the reliability of an embedding-similarity approach.

4. Pending Design Decisions

- Exact size of the curated data subset (the current plan targets roughly 20,000 to 30,000 pairs), balancing embedding-generation time against having enough data for a statistically meaningful per-category breakdown.
- Whether to run the COSMOS dataset as a secondary generalization check, or keep the evaluation scope to NewsCLIPpings only if time becomes limited.
- Whether to add a retrieval-based explainability feature (showing a visually similar genuine pair alongside a flagged result), or keep the interface to a score and verdict only.
- How to handle broken or dead image URLs encountered during acquisition: skip and exclude, retry with backoff, or substitute from an alternate mirror if one exists.
- Where to host the live demo for the final presentation and poster session: running locally on a laptop versus a public deployment on Streamlit Community Cloud or Hugging Face Spaces.

5. Project Goals for Academic Success

This project is designed to demonstrate the full data science lifecycle covered by the program: real data acquisition and validation, data management and preparation, multimodal machine learning analysis, staged model development with rigorous statistical evaluation, and deployment with visualization. It applies coursework in machine learning, statistics, and software development to a well-defined, benchmarked problem, and produces a defensible, evaluated result rather than a purely illustrative demo.

6. Additional Goals for Real-World Application

Beyond the course requirements, the finished system is intended to work as a genuine, usable proof-of-concept: something a fact-checker, journalist, or curious member of the public could actually try, not only read about. The completed code, documentation, and live demo will also serve as a portfolio artifact demonstrating applied multimodal machine learning, and the documented pipeline is written so it could plausibly be extended after the semester, for example into a browser extension or a batch-checking tool for a larger set of images.

7. Updated Project Plan and Timeline

This follows the same 14-week structure submitted with the Project Plan, with Week 1 beginning September 13, 2026. Weeks 1 and 2 are marked complete below; the remaining weeks reflect the design decisions made in Sections 3 and 4 above. Dates remain planning estimates and will be reconciled against the official course calendar as deadlines are confirmed.

Week	Milestone / Deliverable	Activities
1	Project kickoff & refinement	Review the approved proposal; finalize the project objectives and the CLIP-embedding-based detection methodology. (Completed.)
2	Dataset exploration & acquisition	Identify and begin acquiring data from NewsCLIPpings and VisualNews; review COSMOS as a candidate secondary dataset. Project Plan submitted this week. (Completed.)
3	Data preprocessing	Clean, validate, and structure the curated image-caption subset; resolve broken image URLs and set up local dataset storage. Design Document due this week.

Week	Milestone / Deliverable	Activities
4	Exploratory embedding analysis	Generate initial CLIP embeddings; analyze similarity-score distributions and cosine similarity across genuine and falsified pairs.
5	Baseline modeling	Build and evaluate a similarity-threshold baseline classifier for genuine vs. out-of-context pairs.
6	Visualization of categories and trends	Create embedding-space scatter plots, similarity-score distributions, and breakdowns by mismatch-generation method (person-match, scene-match, semantic-match).
7	Midterm progress report	Submit a summary of findings so far; receive instructor feedback on baseline results and modeling direction.
8	Classifier development - Part I	Train and tune a logistic regression classifier on the CLIP embeddings; compare performance to the baseline.
9	Classifier development - Part II	Train and tune a lightweight neural network classifier; perform a per-category performance gap analysis against the logistic regression model.
10	Optional: secondary generalization check	If time allows, evaluate the trained classifier on the COSMOS dataset to test generalization beyond NewsCLIPpings' mismatch-generation methodology.
11	Application interface development	Build out the Streamlit demo interface (image upload, preset examples, results panel) and integrate the best-performing trained classifier for live inference.
12	Report writing & visual storytelling	Draft the final report and documentation; refine the evaluation dashboard and embedding visualizations.
13	Slide deck preparation & report revision	Create the final presentation slides and poster; finalize the written documentation and code repository.
14	Final presentation & submission	Deliver the final presentation and submit the Final Project and Documentation.

8. Risks, Unknowns, and Concerns

- **Link rot in source images:** VisualNews' image URLs point to original news sites and some may no longer resolve. Mitigation: retry logic, generous acquisition buffer above the target subset size, and a fallback to a smaller subset if needed.
- **Compute and time constraints:** generating CLIP embeddings for tens of thousands of images can be slow on a personal laptop CPU. Mitigation: use Google Colab's free GPU tier for this step, and reduce the subset size if it remains a bottleneck.
- **Generalization limits:** NewsCLIPpings' falsified pairs are synthetically generated by caption-swapping, which may not fully represent how out-of-context misinformation looks in the wild. This will be stated explicitly as a limitation rather than implied to be solved.
- **Category difficulty imbalance:** the person-match, scene-match, and semantic-match subsets may have very different difficulty levels, risking a model that looks strong in aggregate but performs poorly on the hardest one. This risk is directly why per-method evaluation was made a first-class design decision rather than an afterthought.
- **Schedule risk from optional features:** the pending decisions around explainability and public hosting are nice-to-have rather than core requirements. If the core pipeline falls behind schedule, these will be the first things deprioritized, given that no late work is accepted in this course.